

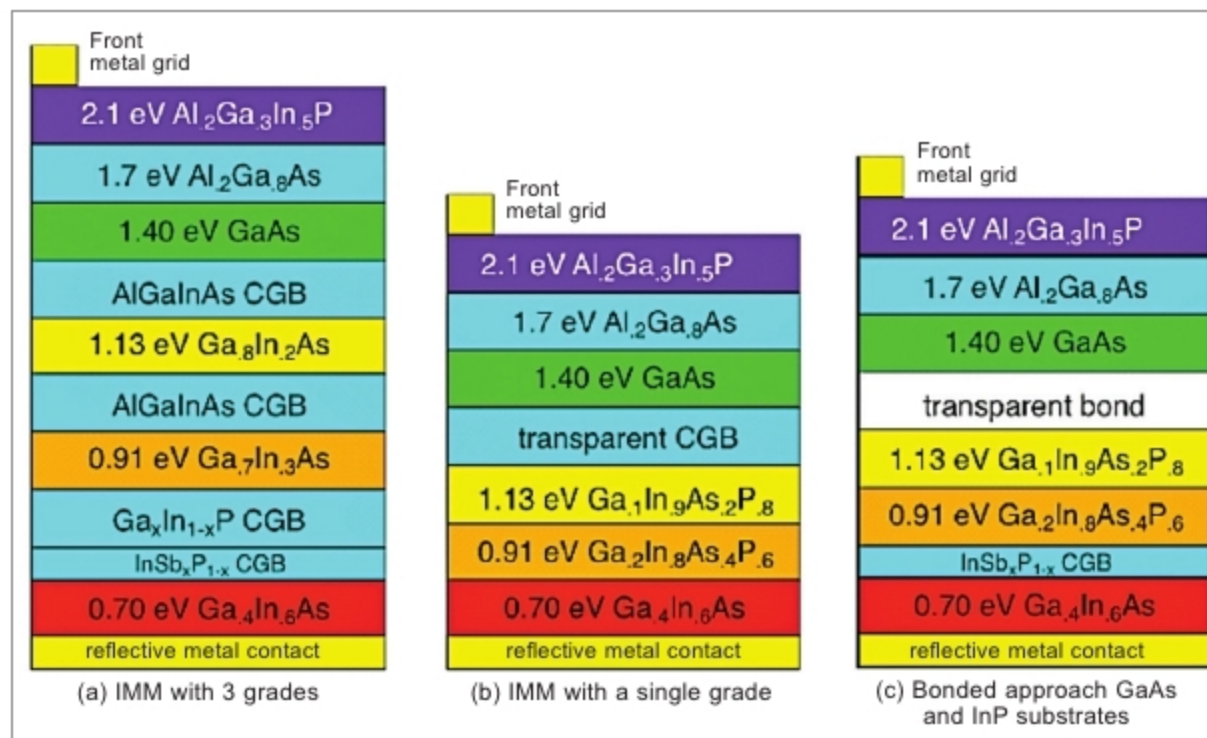


Fig. 8: Anatomy of advanced solar cell with 47 per cent efficiency

ance connection between the p-type back layer of a subcell and the n-type window layer of the subcell beneath it. If tunnel junction is not provided, this p-n junction could produce a photovoltage that could roughly negate photovoltage generated by the top cell.

After every diode layer, a high quality and highly transparent back surface field passivation layer (BSF), typically with a higher band gap than that of the next sublayer cell, is introduced to prevent carrier recombination by repelling them back to the p-n junction without increasing the series resistance of the device. Thus, it increases the efficiency of solar cells. The high recombination velocity at the interface otherwise would affect photoresponse, particularly the red colour and open-circuit voltage to a great extent. AlGaInP and AlInP are popularly used as BSF layers.

Production

Multi-junction III-V cells are assembled in an epitaxial monolithic stack with subcells connected in series through tunnel junctions. Constructing a multi-junction cell in monolithic stack results in material constraints, and fabricating such devices is facilitated if the individual layers of subcells have compatible atomic lattice positions

and are lattice-matched.

This is why Ge, which is lattice-matched to some III-V alloys, is traditionally used as substrate and narrow band gap cell in MJs. Lattice matching limitations can be overcome using wafer bonding or metamorphic buffer layers.

The tunnel-junction layer is constructed by the interphase of highly doped p++ and n++ layers. The interaction of these layers results in a spatially narrow space-charge region, which allows current to flow between subcells. High band gap layers, known as window layers and back surface fields can be added to passivate surface states at the interface between subcell and tunnel junction.

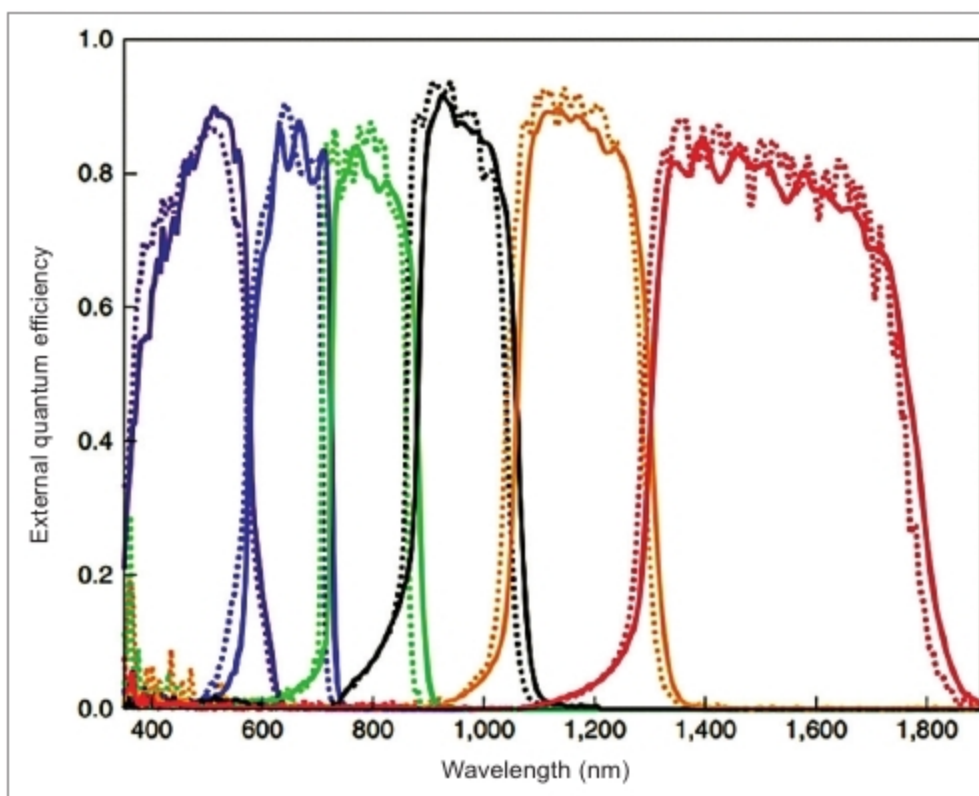


Fig. 9: Solar spectrum utilisation of six-junction advanced solar cell

If the subcells are connected in series, the subcell that conducts the smallest current limits the maximum current that can flow through the device. Therefore a considerable effort is placed on tuning the current of the subcells. Luminescent coupling between subcells can relax some of the current matching design requirements.

MJ III-V solar cells are fabricated commercially in large metal-organic chemical vapour deposition (MOCVD) reactors. Layers can be grown from compounds of Ga and arsine in the hydrogen carrier gas and using dopants such as hydrogen selenide, silane and diethyl zinc.

Advanced MJ solar cells with 47.1 per cent efficiency

This year, in 2020, a group of scientists of NREL, USA, developed a six junction III-V solar cells with a 47.1 per cent conversion efficiency rate under 143 sun concentration and 39.2 per cent efficiency under one sun illumination. The device contains about 140 total layers of various III-V materials, but three times narrower than a human hair.

The new solar cells use a monolithic, series-connected, six-junction inverted metamorphic structure operated under the direct spectrum of the sun. Nearly optimal band gaps for six junctions were fabricated. During fabrication, it was ensured to minimise threading dislocations in lattice-mismatched III-V semiconductors and prevent phase segregation.

Three strategies have been adopted to make these advanced MJ solar cells with extraordinary high efficiency, as shown in Fig. 8, and their solar spectrum utilisation is shown in Fig. 9.

The scientists also developed metamorphic grades from the GaAs to the InP lattice constant, which are transparent to sub-GaAs band gap light. The inverted metamorphic multijunction (IMM) with six junctions demonstrated about fifty per cent efficiency under 1500 suns. **EFY**